

KPI ADVISORY & ANALYTICS

Functional Specification Document

A unified blueprint translating business strategy into governed, measurable performance metrics.

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1. Executive Summary

This document outlines the functional specifications for the approved performance metrics of the organization. Each metric is mapped to strategic objectives and key result areas to ensure operational alignment and governance.

2. KPI Landscape Overview

The following table provides a high-level catalog of all approved performance indicators within the scope of this transformation initiative.

KPI ID	KPI Name	Category	Functional Area
KPI-001	Overall Equipment Effectiveness (OEE)	Operational	Supply Chain

3. Strategic Traceability Matrix

This matrix illustrates the strategic alignment from executive objectives down to specific key performance indicators, providing visibility into strategic translation.

Strategic Objective	Business Challenge	KRA	Functional Area	KPI Name
Reduce machine downtime → High setup times → Operational Excellence → Supply Chain → Overall Equipment Effectiveness (OEE)	High setup times and machine downtime leading to reduced operational efficiency and increased costs.	Operational Excellence	Supply Chain	Overall Equipment Effectiveness (OEE)

4. Individual KPI Specifications

Metric 01: Overall Equipment Effectiveness (OEE)

KPI Category	Operational	Functional Area	Supply Chain
Business Owner	Director of Operations / Head of Clinical Engineering	Data Owner	Operations Analyst / EMR Data Steward

Business Definition

Overall Equipment Effectiveness (OEE) is a best-practice metric that measures the percentage of planned production time that is truly productive. It is calculated by multiplying three key components: Availability Rate, Performance Rate, and Quality Rate. The unit of measure is a percentage (%). The reporting frequency is Daily. The measurement scope is per machine or production line.

Business Purpose & Strategic Relevance

The Overall Equipment Effectiveness (OEE) KPI is fundamental to addressing the strategic priority of 'Reduce Machine Downtime' and the business challenge of 'High Setup Times'. By integrating Availability, Performance, and Quality rates, OEE provides a holistic view of machine productivity, pinpointing specific areas of loss. This metric is crucial for identifying inefficiencies in machine operation, scheduling, and quality control, thereby enabling targeted interventions. For instance, a low Availability rate might signal excessive setup times or unplanned stoppages, directly prompting investigations into process optimization or maintenance schedules. A low Performance rate could indicate machines running slower than their ideal speed, suggesting calibration issues or operator training needs. Finally, a low Quality rate highlights production defects, pointing to process control or material quality issues. The insights derived from OEE directly support management decisions regarding resource allocation for preventative maintenance, investment in automation to reduce manual setup, and the implementation of standardized operating procedures to minimize variability. Without monitoring OEE, the organization would lack a quantifiable measure of its core operational efficiency, making it difficult to prioritize improvement initiatives, justify capital expenditures for equipment upgrades, or effectively track progress against the 'Asset Productivity' KRA. This KPI is essential for driving operational excellence and ensuring that critical healthcare equipment is utilized to its maximum potential, directly impacting service delivery and cost management.

Calculation Methodology

Formula	Availability Rate * Performance Rate * Quality Rate
Numerator Details	The numerator is derived from the product of (Operating Time * Actual Throughput * Good Units). Operating Time is defined as Planned Production Time minus Downtime. Actual Throughput is the actual rate at which units are produced. Good Units are the units produced that meet quality standards.
Denominator Details	The denominator is Planned Production Time multiplied by the Ideal Run Rate and Total Units. Planned Production Time is the total scheduled operating time for the equipment, excluding planned breaks and scheduled maintenance. Ideal Run Rate is the theoretical maximum production speed of the equipment. Total Units are all units produced, including those that do not meet quality standards.
Calculation Logic	1. Determine Planned Production Time for the period. 2. Record all instances and durations of Downtime (unplanned stops). 3. Calculate Operating Time: Planned Production Time - Downtime. 4. Measure Actual Throughput (units produced per unit of time) during Operating Time. 5. Record Total Units produced. 6. Record Good Units produced (meeting quality standards). 7. Calculate Availability Rate: Operating Time / Planned Production Time. 8. Calculate Performance Rate: (Ideal Run Rate * Total Units) / Operating Time. 9. Calculate Quality Rate: Good Units / Total Units. 10. Calculate OEE: Availability Rate * Performance Rate * Quality Rate.
Inclusion Rules	Include all production runs during scheduled operating hours. Include all units produced, regardless of quality status, for the Performance and Quality Rate calculations. Include all unplanned downtime events.
Exclusion Rules	Exclude planned maintenance periods from Planned Production Time. Exclude scheduled breaks (e.g., lunch, shift changes) from Planned Production Time. Exclude units produced outside of scheduled operating hours.
Sample Worked Example	Assume a machine is scheduled to run for an 8-hour shift (480 minutes). Planned Production Time = 480 minutes. During the shift, there was 30 minutes of unplanned downtime (e.g., machine jam). Downtime = 30 minutes. Operating Time = 480 minutes - 30 minutes = 450 minutes. Assume the machine produced 900 units in total, with an ideal run rate of 2 units per minute. Total Units = 900 units. Assume 850 of these units met quality standards. Good Units = 850 units. Ideal Run Rate = 2 units/minute. 1. Availability Rate = Operating Time / Planned Production Time = 450 / 480 = 0.9375 (93.75%) 2. Performance Rate = (Ideal Run Rate * Total Units) / Operating Time = (2 units/min * 900 units) / 450 minutes = 1800 / 450 = 4.0. *Correction: The formula for Performance Rate should be: (Total Units / Operating Time) / Ideal Run Rate. Let's re-calculate with correct logic: Performance Rate = (900 units / 450 minutes) / 2 units/min = 2 units/min / 2 units/min = 1.0 (100%). This implies the machine ran at its ideal speed when it was running. 3. Quality Rate = Good Units / Total Units = 850 / 900 = 0.9444 (94.44%) OEE = Availability Rate * Performance Rate * Quality Rate = 0.9375 * 1.0 * 0.9444 = 0.8859 Therefore, OEE = 88.59%

Business Rules & Data Validation

Business Operating Rules	Planned Production Time is defined as the total time the equipment is scheduled to operate, excluding scheduled breaks and planned maintenance. Downtime refers strictly to unplanned stoppages that halt production. The Ideal Run Rate is a pre-defined technical specification of the machine's maximum output capacity under ideal conditions. Good Units are those that pass all quality inspections.
Data Validation Rules	Availability Rate, Performance Rate, and Quality Rate must each be between 0% and 100%. OEE must be between 0% and 100%. Total Units must be greater than or equal to Good Units. Operating Time must be less than or equal to Planned Production Time. Downtime must be non-negative.
Exception Handling Rules	If Planned Production Time is zero, OEE is undefined (or reported as 0% with a note). If Operating Time is zero, Availability Rate is 0%, and OEE is 0%. If Total Units is zero, Quality Rate is undefined (or reported as 0% with a note), and OEE is 0%. If Ideal Run Rate is zero, Performance Rate is undefined (or reported as 0% with a note), and OEE is 0%. Handle null values by treating them as zero for calculation purposes, but flag for data quality investigation.
Data Quality Expectations	Source data for downtime, units produced, and quality status must be captured in near real-time. Data completeness for all components (Planned Production Time, Downtime, Total Units, Good Units) should be at least 99%. Data accuracy targets should be established based on source system capabilities and business process controls.

Recommended Source Systems

Recommended Source Systems: Manufacturing Execution System (MES) for real-time production data (units, run rates, downtime events), ERP Finance/Controlling module for planned production schedules and potentially cost-related data. Recommended ERP Finance Ledger for financial reporting alignment.

5. Governance Framework

To ensure metric consistency, accountability, and ongoing relevance, a formal governance framework is established for all approved KPIs. This structure assigns clear responsibilities and defines escalation paths.

Roles and Responsibilities

- **Business Owner:** Responsible for defining the business logic, validating calculation results, approving target thresholds, and driving operational performance based on metric insights.
- **Data Owner:** Accountable for technical lineage, data completeness, source-to-target mapping, ETL data quality checks, and resolving data ingestion or availability issues.
- **Escalation Path:** In case of data quality discrepancies or alignment disputes, issues are escalated to the Data Governance Committee and KPI Advisory Board for review and reconciliation.

Ownership and Governance Policy Matrix

Metric Name	Business Owner	Data Owner	Governance Policy Notes
Overall Equipment Effectiveness (OEE)	Director of Operations / Head of Clinical Engineering	Operations Analyst / EMR Data Steward	The Director of Operations / Head of Clinical Engineering is the business owner responsible for performance against OEE targets. The Operations Analyst / EMR Data Steward is the data owner responsible for the accuracy and availability of the underlying data. OEE performance will be reviewed weekly by the Operations Management team and monthly by the Steering Committee. Variances exceeding 5% from target will require immediate investigation and documented action plans.

6. Reporting & Dashboard Requirements

Visualization layout and threshold criteria dictate how data is displayed to support decision-making. The matrix below outlines reporting recommendations and performance thresholds for each metric.

Metric Name	Reporting Guidelines	Dashboard Placement	Threshold Guidance
Overall Equipment Effectiveness (OEE)	Trend line chart showing OEE over time (daily, weekly, monthly). Stacked bar chart breaking down OEE into Availability, Performance, and Quality components. Drill-down dimensions: by machine, by production line, by shift, by reason code for downtime, by reason code for quality defects. Alerting rules: Trigger alerts if OEE drops below 70% (Red threshold) or if any component (Availability, Performance, Quality) drops below its respective Red threshold. Threshold levels: Green >= 75%, Amber 70-75%, Red < 70%.	Operations Control Console, Plant Performance Dashboard, Executive Performance Dashboard (summary view).	Green: Consistently above 75%, indicating efficient operations with minor losses. Amber: Between 70% and 75%, signaling a need for focused improvement efforts. Red: Below 70%, requiring immediate root cause analysis and corrective actions. Targets should be set based on historical performance, industry benchmarks, and strategic goals.

7. Assumptions & Constraints

A clear understanding of business and technical assumptions is critical for successful implementation. The following list represents consolidated baseline assumptions and constraints.

General Architectural Assumptions

- **Data Availability:** Source transactional tables are assumed to be loaded into the central reporting repository on a standard nightly batch cadence.
- **Fiscal Calendar:** Standard calendar year rules are assumed unless otherwise explicitly documented in specific financial indicators.
- **System Uptime:** Target source ERP ledgers are assumed to maintain 99.5% uptime during standard reporting extraction windows.

Metric-Specific Assumptions and Limitations

- **Overall Equipment Effectiveness (OEE):** Assumes that the 'Ideal Run Rate' for each piece of equipment is accurately defined and maintained. Assumes that downtime is accurately recorded in the source system. Assumes that the distinction between planned and unplanned downtime is consistently applied. Assumes that quality inspection data is reliably captured. Constraint: OEE calculation is sensitive to the accuracy of its components; poor data quality in any one area will significantly impact the overall metric.

8. Implementation Considerations

Transitioning these specifications into functional BI tools requires rigorous testing, change management, and data reconciliation. Standard implementation considerations are outlined below.

Standard Implementation Guidelines

- **Data Reconciliation:** All KPI calculation outcomes must be audited and reconciled against official books of record or audited financial statements.
- **User Acceptance Testing (UAT):** Business owners must perform visual and numeric validation of dashboard mockups prior to production sign-off.
- **Change Management:** Training sessions and clear system documentation are required to support user onboarding and ensure high organizational adoption.

Metric-Specific Technical Considerations

- **Overall Equipment Effectiveness (OEE):** Prioritize the accurate definition and consistent recording of 'Downtime' and 'Good Units' as these are often sources of data quality issues. Implement standardized procedures for machine setup and quality checks to improve Availability and Quality components. Conduct operator training focused on understanding OEE and their role in improving it. Establish a cross-functional team (Operations, Maintenance, Quality) to review OEE performance and drive improvement initiatives. Traceable to alignment meeting decision on prioritizing operational efficiency metrics. The roles of Operations Analyst and EMR Data Steward should be formally assigned and trained on OEE data requirements and validation.

9. Appendix

KPI Glossary

- **KPI (Key Performance Indicator):** A quantifiable measure used to evaluate the success of an organization or activity in meeting performance objectives.
- **KRA (Key Result Area):** Primary focus areas of outcomes or outputs for which an organizational unit or role is responsible.
- **Numerator:** The upper portion of a division representing the measured subset of occurrences.
- **Denominator:** The lower portion of a division representing the total base population.

Data Quality Principles

- **Accuracy:** Data correctly represents the real-world operational event it records.
- **Completeness:** All necessary dataset components are present without omission.
- **Consistency:** Metrics align across various systems, business units, and report interfaces.
- **Timeliness:** Updates occur within the required reporting cadence and operational windows.

Acronym Reference

- **ERP:** Enterprise Resource Planning
- **SAP FI-CO:** Financial Accounting & Controlling
- **SAP SD:** Sales & Distribution
- **SAP MM:** Materials Management
- **BI / DWH:** Business Intelligence / Data Warehouse
- **UAT:** User Acceptance Testing